Green Scores: Using Machine Learning to Predict Renewable Energy Favorability for Various Global Cities

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***Abstract*– An important aspect to reducing global greenhouse gas emissions is the transition to the use of renewable energy sources. This transition necessitates the ability to predict whether a city is capable of harnessing solar or wind power, in order to successfully establish infrastructure and ensure a stable grid. This project attempts to predict a custom-calculated "green score" for ten global megacities, which represents the combined potential for solar and wind energy generation.**

We accomplish this by using a year's worth of hourly time-stamped observations from 2025, including shortwave radiation, wind speed, latitude, and longitude. Additional environmental data such as cloud cover, temperature, and relative humidity– all of which can affect the actual amount of energy generated– will also be aggregated. The latter makes the green score prediction much more robust and meaningful for real-world application.

Here, the project's goal is to use these atmospheric observations to train a machine learning algorithm to predict a target label– green score– for various cities and then apply the trained model on new weather data from Charleston, SC. In doing so, this project will provide a data-focused framework for identifying cities with high renewable favorability, aiding in the transition to clean energy.

The models will be trained and tested on 1,138,800 data points (in a traditional 80/20 split) collected from the OpenMeteo API. Feature engineering on the timestamp variable will be necessary and decomposed into years, months, days, and hours for use. A correlation heatmap will be used to observe any meaningful relationships between the variables and the target label.

In order to evaluate the results of the models, we will perform five-fold cross-validation. This project will be implemented in Google Colab Notebook, using Pandas for dataframe manipulation, NumPy for numerical computation, Matplotlib and Seaborn for visualization, and Scikit-learn for machine learning techniques.

After confirming the best algorithm, we will then use the trained model on unseen data in order to predict its own green scores and assess whether that city is suitable for renewable energy infrastructure.

I. INTRODUCTION

In order to tackle the problem of rising global greenhouse gas emissions, we need to be able to identify cities that are capable of sustaining renewable energy solutions such as solar panels and wind turbines, among others outside of the scope of this project.

Switching from the overconsumption of fossil fuels to clean energy is important because these emissions are rapidly changing conditions on Earth– through rising sea levels, increased intensity and frequency of natural disasters, endangering human health, damaging ecosystems, reducing agricultural yields, and shifting economies.

Assessing renewable favorability requires a metric that evaluates how "good" climatic conditions are throughout the day for renewable energy generation, which in this project is called "green score." This score will be re-engineered from a Kaggle dataset called "Global Green Energy Pulse: Real-Time Renewable," which was created by Nudrat Abbas for comparative analysis and machine learning. It is a continuous variable with values ranging from 0 to 1, and is formulated through minimum-maximum Normalization on the shortwave radiation and wind speed variables.

We will specifically be using hourly time-stamped data from the entire year of 2025 from ten global megacities– New York, Tokyo, London, Mumbai, Sydney, Berlin, Sao Paolo, Cairo, Lagos, and Beijing. Since most of these cities are located in the Northern Hemisphere and two are located in the Southern Hemisphere, we will analyze whether there is a difference in the distribution of green scores.

We will visualize how these environmental observations change throughout the day and observe seasonal trends. We will attempt to rule out any features in the dataset that possess no correlation to our target label or risk introducing redundancy when training the models.

Since we are trying to predict a continuous variable, we will implement supervised machine learning algorithms, including Linear Regression, Decision Tree Regressor, and the ensemble Random Forest Regressor method. Upon analysis of these models and their cross-validation scores, we will choose the best one and apply it to unseen weather data from Charleston SC in order to assess whether or not it is suitable for renewable energy infrastructure.

II. DATA COLLECTION & PREPROCESSING

The data for this project was collected using a python API call from OpenMeteo– an open-source weather API that aggregates data from national weather services, such as NOAA, and other global meteorological centers. Their aim is to provide free and easily accessible weather data, both historical and forecasting, for non-commercial use. We returned to this method of collecting data at the end of the project by performing a second API call in order to apply the trained machine learning model to previously unseen data.

Since this project involved recreating the aforementioned Kaggle dataset, we decided to import it into the Notebook for reference by using the kagglehub library and reading the CSV file in that way.

III. DATA PREPROCESSING

The processing portion of this project took us quite some time, as we did not have past experiences in performing API calls and subsequently parsing the data from its JSON object formatting to a pandas DataFrame.

In order to recreate the Kaggle “Global Green…” dataset, the resulting API call included the latitude and longitude for each of the global cities, followed by appending the start_date as 2025-01-01, the end_date as 2025-12-31, and the hourly call for the following variables: temperature_2m, relative_humidity_2m, cloud_cover, wind_speed_100m, and shortwave_radiation.

Using a for loop, that data was then parsed into a pandas DataFrame. Because each city name was not readily available in OpenMeteo, we merged the “Global Green…” dataset’s “city” column on “latitude” and “longitude” between both (the original and our new) datasets to create that column, for investigative use.

Once our dataset was successfully recreated in its initial stage, we checked for null values and found that there were none. Therefore, we did not need to implement any missing value imputation methods such as interpolation, which would have been suitable for this time-series data.

We then moved on to recreating the green_score feature using the following formula:

```
normalized_value = (x – min) / (max – min)
green_score = 0.5 * solar_normalized + 0.5 * wind_normalized
```

This gave us a new column in the dataset containing floating point values ranging from 0 to 1. Analysis of this new column showed us that for all ten cities combined, the minimum value was 0.0000, the maximum value was 0.8732, the mean was 0.2114, and the median was 0.1751. These values will later become important when using the trained model on the unseen weather data.

Lastly, before we moved on to visualizations, we converted the “time” column using the pandas to_datetime() function, and then aggregated columns for year, month, day, and hour and dropped the initial time column. This made further analysis of the weather variables by the different time factors easier. After completing this preprocessing, we ended up with a DataFrame consisting of 87,600 rows and 13 columns for a total of 1,138,800 data points.

IV. METHODOLOGY

A. Exploratory Data Analysis

The first step in the investigation of our data was to perform a correlation analysis. This would help us in confirming which features would be necessary and unnecessary in predicting the target green score label.

As shown in Figure 1, the features with the highest correlations to green score were shortwave radiation (+0.87), wind speed (+0.38), temperature (+0.29), and relative humidity (-0.47). The resulting correlations show us that the previous features will be

essential when training the machine learning model and that the latitude, longitude, and cloud cover variables hold no predictive power in determining green score.

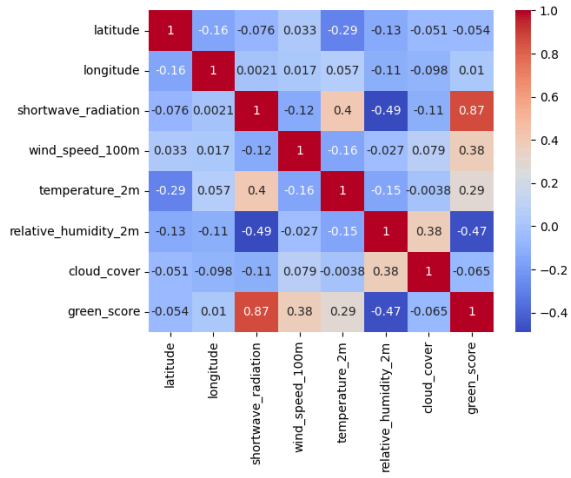

Figure 1. Correlation heatmap of the continuous quantitative variables, excluding city (categorical), and year, month, day, and hour (independent indices).

It is also important to note the other relationships which exist among the features. Namely, temperature and shortwave radiation (+0.4), shortwave radiation and relative humidity (-0.49), and relative humidity and cloud cover (0.38). For successful model training, we will need to eliminate variables that may lead to multicollinearity or redundancy issues in the model, and so we will possibly need to eliminate temperature based on this information.

We then wanted to further investigate the redundant relationship between temperature and shortwave radiation, so we created two side-by-side subplots that show the averages by hour for all the cities in the dataset, as shown in Figure 2. Using this visualization, we can confirm that these variables move together throughout the day and therefore we really only need one to train the model.

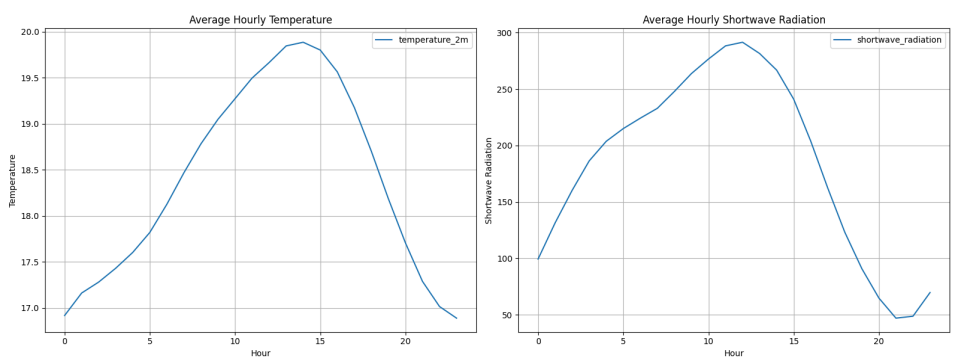

Figure 2. Side by side lineplots of Average Hourly Temperature and Average Hourly Shortwave Radiation for all 10 megacities.

In terms of what this means for our analysis, we can observe a strong diurnal trend for both variables such that temperature and shortwave radiation are low in the morning, rise throughout the day and come to a peak around noon or soon thereafter, decrease in the afternoon, and reach another low at the end of the day. This trend indicates optimal times in the middle of the day for solar panel energy generation and suggests which time of day would show higher green scores.

In the same fashion that we explored the temperature and shortwave radiation variables, we also investigated how the relative humidity (Figure 3) and cloud cover (Figure 4) variables move throughout the day.

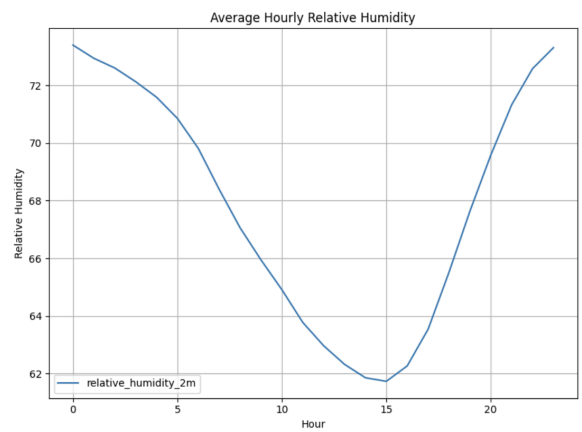

Figure 3. Average hourly relative humidity for all 10 cities.

The movement of relative humidity throughout the day is interesting because it showcases an inverse relationship between it and temperature. Relative humidity tends to start at a higher value in the early hours of the day, decrease as temperatures rise throughout the day, and increase again as temperatures cool towards the end of the day. This trend is important because humidity in the air causes water droplets to form and scatter on solar panels, which absorb sunlight, thus limiting how much energy is generated through these systems.

Cloud cover, on the other hand, is much more erratic in the middle of the day with no obvious trend. However, the steady rise in the early hours and

the steady decline towards the end of the day signal a trend related to solar heating– in order for common convective clouds to form, solar radiation is needed, which increases as the sun rises and decreases as it sets.

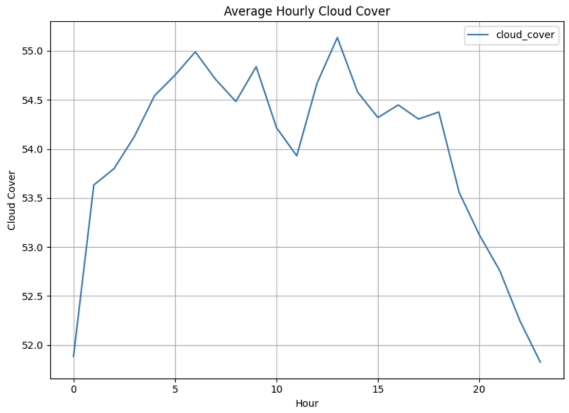

Figure 4. Average hourly cloud cover for all 10 cities.

After analyzing these features, we proceeded to explore the green score variable with bar plots and histograms in order to understand the distribution amongst the cities.

First, we plotted the distributions of green scores for cities in the Northern Hemisphere (Figure 5) and another for the cities in the Southern Hemisphere (Figure 6). We did this in order to retain the seasonal variability experienced in each hemisphere.

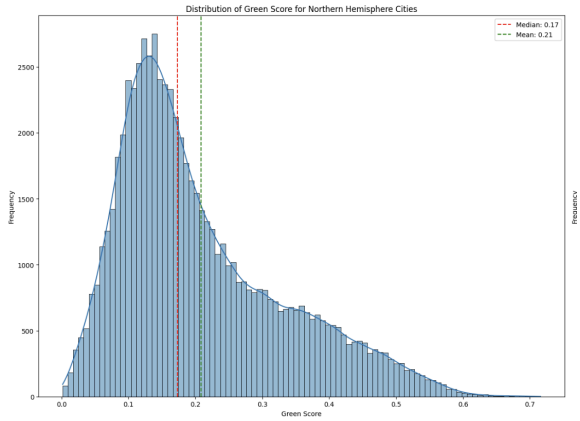

Figure 5. Histogram for the distribution of green scores for the cities in the Northern Hemisphere (latitude >= 0).

For cities in the Northern Hemisphere, we found that the median score was 0.17 and the average score was 0.21. For cities in the Southern Hemisphere, the median and mean both increased to 0.19 and 0.22, respectively. These graphs and their labeled data will become more important as we predict green scores using the trained model.

The histogram in Figure 7 is a decomposed display of the averages of green scores for each city, ranked in descending order.

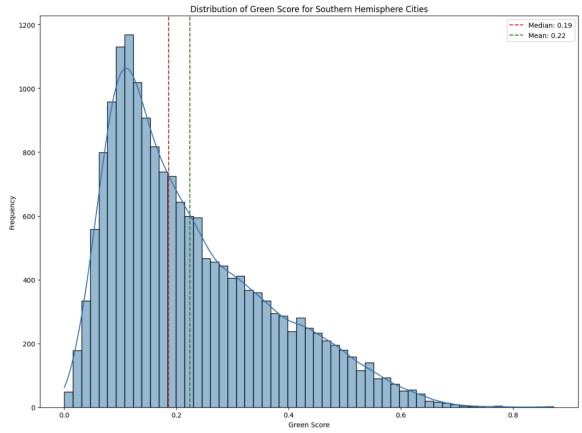

Figure 6. Histogram for the distribution of green scores for cities in the Southern Hemisphere (latitude < 0).

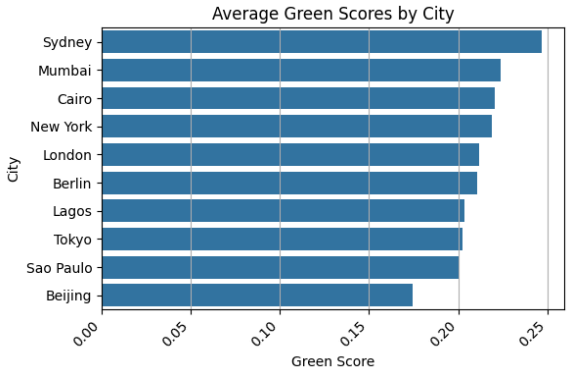

Figure 7. Horizontal bar plot displaying the average green scores for each of the ten cities in the dataset.

` The cities with the highest average green scores are Sydney, Mumbai, and Cairo, with average scores ranging from 0.220 to 0.247, as these are all generally sunnier cities. The cities with the lowest scores are Tokyo, Sao Paolo, and Beijing, with average scores ranging from 0.174 to 0.203, with Beijing being the only city with an average score under 0.2. This could be influenced by known high air pollution and low air quality.

B. Model Pipeline

In order to find a suitable model, we chose to train the following three: Decision Tree Regressor, Linear Regression, and Random Forest Regressor. We felt these were suitable because of the nature of our time-series dataset and because we are predicting a continuous target variable.

Specifically, for the Decision Tree algorithm, this model would be ideal had we not known the relationships that exist in our dataset. If the formula for the green scores was not available to us, we

would not know whether linear or non-linear relationships exist. By training and testing with this algorithm, we can begin to understand the complex relationships at play.

We also chose to test the Linear Regression algorithm because of the obvious relationships between shortwave radiation, wind speed, and the engineered green score feature. In addition, it is a generally a good base model to test when predicting a continuous variable.

For our third algorithm choice, we decided to use the Random Forest Regressor ensemble learner because it mitigates any overfitting that may present in only using Decision Tree, is much more robust for complex non-linear relationships, and is also able to analyze feature importance.

It is important to note that we tested the models without dropping any features first and then proceeded to eliminate the temperature and cloud cover features according to our previous analysis. In this step, we also eliminated latitude and longitude in favor of boolean values for city names by using the pandas get_dummies() function.

For all models, we chose the traditional 80/20 train/test split and evaluated performance based on the traditional metrics– the coefficient of determination (R^2) values and MAEs (Mean Absolute Errors).

C. Model Validation

In order to validate the performances of all three models, we relied on five-fold cross validation to provide an equal chance for all of the data to be used for training. This creates a much more reliable estimate of how the models perform with unseen data.

	Model	MAE	MSE	RMSE	R2 Score	Cross-Validation R2 Score
0	Decision Tree Regressor	0.0010	0.0000	0.0025	0.9996	0.9997
1	Linear Regression	0.0000	0.0000	0.0000	1.0000	1.0000
2	Random Forest Regressor	0.0004	0.0000	0.0012	0.9999	0.9999

Figure 8. Table showing final results of the performance of all three regression models (Decision Tree, Linear Regression, and Random Forest Regressor), after dropping features.

V. RESULTS

A. Regression Models

The first model we trained and tested was the Decision Tree algorithm– before and after eliminating non-correlated features. There was no change in the determination coefficients, but there were slight changes in the error scores. We received an R^2 of 0.9996 and MAEs of 0.0011 (before dropping temperature and cloud cover) and 0.0010 (after dropping). This lowering in error, although slight, confirmed our observations in the initial data analysis and allowed us to proceed in training the next two models without these features, as shown by the results in Figure 8.

The R^2 value for the Linear Regression model proved to be the best outcome with a perfect score of 1 and an error score of 0. We will later discuss why this is the case, but for now we will accept that this scenario rarely, if ever, occurs in real-world data-driven investigations.

For the ensemble learner, we received an R^2 of 0.9999 and an MAE of 0.0004, which outperformed the single Decision Tree model, successfully capturing the complexity of the data. For both of these models, however, they were both able to equally identify the features of highest importance, as shown in Figures 9 and 10, with shortwave radiation having the most predictive power (roughly 75%), followed by wind speed (roughly 25%).

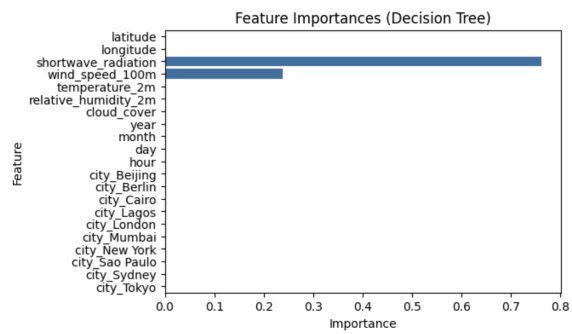

Figure 9. Decision Tree Feature Importances.

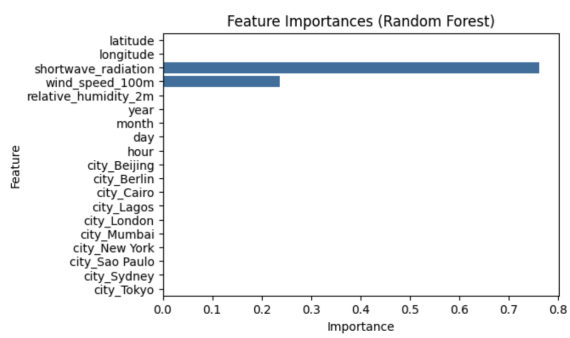

Figure 10. Random Forest Feature Importances.

A. Model Application

Our next steps were to apply the successfully trained model to more unseen data. We performed another API call to OpenMeteo to gather hourly weather data from the entire year of 2025. We created a new DataFrame with the data, and applied

the same configurations as we did for the initial dataset.

We proceeded to use the trained Linear Regression model to predict green scores for the new data, as shown in Figure 11. Based on the analysis of the predicted scores, the median score was found to be 0.23,

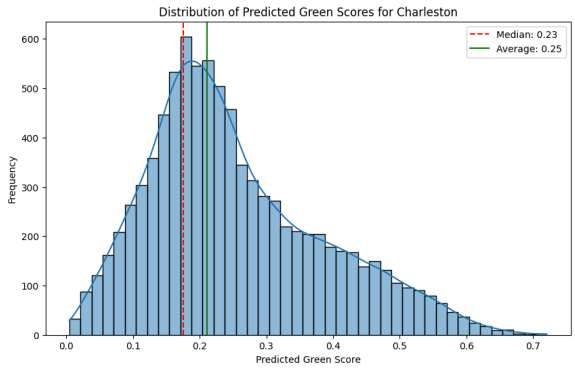

Figure 11. Histogram showing the distribution of predicted green scores for hourly weather data from Charleston SC (2025).

and the average score was 0.25– beating out the averages for all the cities, regardless of whether they were located in the Northern or Southern Hemispheres, suggesting that Charleston is a very ideal candidate for renewable energy infrastructure.

VI. DISCUSSION

The extremely high determination coefficients and extremely low error scores resulting in this project are generally not the norm in real-world data science investigations. These values can be attributed to the way the green score variable was engineered– through the minimum-maximum Normalization of two features of the dataset, which were then combined. This created a linear relationship between each of those variables to the green scores, which explains why the linear regression model worked so well in predicting the target variable.

The Decision Tree and Random Forest models were also extremely successful in predicting the target label, but their performance differed in being able to analyze feature importance. Because they both relied heavily on only shortwave radiation and wind speed, these models essentially treat the other variables as irrelevant and ignore them.

Furthermore, the dataset and these models possess many limitations. When assessing the renewable favorability of a city, there are many more complexities to take into consideration, such as financial burden, land availability, infrastructure readiness, current energy usage, grid capacity, and ambitious policies.

Additionally, we would have liked to have worked with an official sustainability metric, rather than reconstructing one. This would make our findings more robust and meaningful. However, with this methodology, we were able to find some interesting and meaningful revelations about the climatic conditions in Charleston. A similar project with thesel modifications would make for interesting future work.

VII. RELATED WORK

We were not able to find work being done with sustainability metrics, but there is plenty of work being done surrounding renewable energy prediction using similar machine learning techniques.

For example, a Kaggle dataset containing energy grid data from India and China was used in a project to predict energy usage with Random Forest [1]. This project was the closest one we found to ours. Another paper predicted global energy usage and price per share using real-world data [2]. Using data from China, another project centered around predicting three variables: energy usage, fossil fuel consumption, and electricity consumption [3].

Another paper of interest was day-ahead prediction of renewable energy generation using XGBoost [7], which could be applied similarly to the data used in this project.

In a more recent paper published in April 2025, the researchers were able to maximize forecasting performance of renewable energy by incorporating the Ninja Optimization algorithm with a quantum temporal model, resulting in a higher R^2 and lower error [10].

VIII. CONCLUSION

Being able to predict how favorable conditions are for renewable energy generation, as shown in this project, is important to aid in the transition to cleaner energy solutions, such as wind turbines and solar panels.

We were able to use environmental data to train a Linear Regression model to predict green scores, and successfully applied the trained model to unseen data from Charleston SC to predict its own scores. Based on those results, we found that Charleston is a very good candidate city for clean energy solutions. However, we acknowledge that there is so much more work that can be done with this project in the future.

We can apply this template to many cities all around the world in order to assess their capability for this type of infrastructure, in order to provide

policymakers with meaningful insights in order to tackle issues like climate change.

IX. ACKNOWLEDGEMENTS

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X. RELATED WORKS

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